

TASKFLOW USER MANUAL

Premium Project Management & Boundary-Isolated Workspace

This comprehensive document details the setup, architectural elements, user interface components, and security policies of the TaskFlow application.

1. Introduction to TaskFlow

TaskFlow is a premium, beautifully crafted project and team task management application designed for modern workspaces. Featuring fluid interactive Kanban boards, rich dashboard analytics, collaborative comment threads, and deep filtering mechanics, TaskFlow provides an all-in-one workspace for teams to coordinate tasks efficiently and securely.

Built with React 19, Vite, Tailwind CSS, Motion (Framer Motion), Node.js, Express, and Prisma/SQLite, TaskFlow merges peak performance with state-of-the-art interactive user experience. Features are isolated by strict Role-Based Access Control (RBAC), ensuring that team members only access projects they belong to.

2. Getting Started & Installation

TaskFlow can be set up in two ways depending on your preferred environment: a standard local Node.js environment or a containerized Docker deployment.

Prerequisites: Ensure you have Node.js (v18+) and npm/yarn installed, or Docker and Docker Compose configured on your system.

- **Standard Setup:** Clone the repository, run 'npm install' in the project root, duplicate '.env.example' to '.env.local' and add your Gemini API Key.
- **Database Initialization:** Execute 'npm run db:migrate' to apply database schemas and run 'npm run db:seed' to seed the environment with demo data.
- **Execution:** Run 'npm run dev' to concurrently launch the Vite development server (port 3000) and the Express server (port 3001).
- **Docker Deployment:** Alternatively, simply run 'docker-compose up -d --build' to automatically build the project, migrate/seed SQLite, and expose it at <http://localhost:3050>.

3. Authentication & User Security

Security is a core pillar of the TaskFlow platform. All interactions start with a secure login or account registration flow.

Security features include:

- **Brute-Force Protection:** A custom sliding-window rate limiter restricts authentication requests to 10 attempts per 15 minutes per IP address, returning '429 Too Many Requests' if exceeded.
- **Secure Handshakes:** Sessions are secured using JWT-based token handshakes, with custom HTTP security headers (CORS checking, CSP, and Frame Options 'DENY') preventing clickjacking and mime-spoofing.
- **Demo Credentials:** For local test setups, standard seeded users default to the password 'password123'.

4. Dashboard & Workspace Analytics

The Dashboard serves as the central command center for users, showing relevant summaries and action items tailored specifically to the logged-in user's project memberships.

Key elements on the Dashboard:

- Real-time KPIs: Displays total active projects, total tasks, and completed tasks percentage.
- Overdue Indicators: Alerts users to tasks that have passed their deadline.
- Starred Projects: Provides immediate shortcuts to starred or favorite workspaces.
- Upcoming Tasks: Lists the upcoming deadlines sorted chronologically.
- Activity Stream: Shows chronological project activity feeds indicating updates, comments, and milestones.

5. Projects & Interactive Kanban Boards

Workspaces are divided into Projects. Users can create, star, and manage projects from the main index.

Each Project workspace features a premium glassmorphic Kanban Board:

Status Columns: Tasks are organized into 'Todo', 'In Progress', 'In Review', and 'Done' statuses.

Drag & Drop Updates: Users can dynamically drag tasks between columns, triggering smooth Framer Motion transitions and saving updates instantly to the database.

Task Cards: Cards display titles, priority badges (Low, Medium, High, Urgent), due dates, assigned members, and custom tags/labels.

6. Task Details & Team Collaboration

Clicking a Task opens a detailed card overlay where team members can collaborate and manage task attributes.

Collaboration features include:

- Description & Labels: View or edit rich descriptions and assign dynamic colored labels/tags.
- Task Assignments: Reassign tasks to other project members from a dropdown menu.
- Priority & Due Date: Adjust severity and target dates.
- Comment Threads: Post comments inside the task. Comment feeds are chronologically arranged with user avatar initials and timestamps, keeping threads context-bound to individual items.

7. Boundary Isolation & Access Rules (RBAC)

To safeguard workspace confidentiality, TaskFlow implements a rigid Role-Based Access Control (RBAC) model. All operations are strictly bound by project membership.

Enforcement Policies:

- Project Isolation: Users cannot see or fetch details for projects they aren't a member of. Attempting to access an isolated project returns '403 Forbidden'.
- Task Operations: Users can only create, edit, change status, or delete tasks in projects where they hold active membership.
- Collaboration Isolation: Comments can only be read or posted by verified project members.
- Global Search: Search queries only return matches from projects the searching user is explicitly a member of.

8. Troubleshooting & FAQs

Below are common issues and solutions encountered during development or deployment:

- Database Schema Out of Sync: Run 'npx prisma db push' or 'npm run db:migrate' to restore schema integrity.
- Port 3000/3001 Busy: Ensure other local web applications are stopped, or configure alternative port variables in '.env.local'.
- Docker Build Fails: Verify your Docker Daemon is running, and perform a clean build using 'docker-compose down -v' followed by 'docker-compose build --no-cache'.